



西南科技大学

SOUTHWEST UNIVERSITY OF SCIENCE AND TECHNOLOGY

课程主题研究报告

RESEARCH REPORT

Topic: Data Lake vs. Data Warehouse: Architecture and Use Cases

学院名称/School	School of Computer Science and Technology
学生姓名/Name	NOURDINE MOHAMED
学号/STUDENT ID	4420230912
课程/SUBJECT NAME	BIG DATA MANAGEMENT TECHNOLOGY
指导教师/TEACHER NAME	MEMON MUHAMMAD HAMMAD郝礼德
提交日期/DATE	2026-06-10

ABSTRACT

The increasing volume of data generated by businesses, social media platforms, IoT devices, and digital applications has created a growing need for efficient data storage and management solutions. Data Lakes and Data Warehouses are two widely used architectures that help organizations store, process, and analyze large amounts of data. This report examines the architectural differences, advantages, limitations, and practical use cases of Data Lakes and Data Warehouses. The primary objective of the study is to compare these technologies and evaluate their suitability for different business and analytical requirements.

The research is based on a review of recent academic literature, industry reports, and technical publications from the last five years. The findings indicate that Data Lakes provide a flexible and scalable environment for storing structured, semi-structured, and unstructured data in its raw format, making them suitable for big data analytics, machine learning, and artificial intelligence applications. In contrast, Data Warehouses are designed for storing structured and processed data, offering high performance for business intelligence, reporting, and decision-making processes.

The study concludes that both Data Lakes and Data Warehouses play important roles in modern data management. While Data Lakes support advanced analytics and large-scale data storage, Data Warehouses provide reliable and efficient solutions for structured reporting and business insights. The choice between these technologies depends on organizational goals, data types, and analytical requirements. Many organizations are increasingly adopting hybrid architectures to benefit from the strengths of both approaches.

Keywords

Data Lake, Data Warehouse, Big Data, Data Management, Data Analytics, Business Intelligence (BI), Machine Learning, Artificial Intelligence (AI), Structured Data, Unstructured Data, Hybrid Architecture.

REFERENCES

- [1] Amazon Web Services. (2024). *Data lake versus data warehouse: Understanding the differences and use cases*. AWS Documentation.
- [2] Databricks. (2023). *The data lakehouse architecture: Combining the best of data lakes and data warehouses*. Databricks.
- [3] Gartner. (2023). *Modern data management strategies for enterprise analytics*. Gartner Research.
- [4] IBM. (2025). *Modern data architectures for AI, analytics, and business intelligence*. IBM Corporation.
- [5] Microsoft. (2024). *Azure data lake storage and Azure Synapse Analytics architecture guide*. Microsoft Learn.
- [6] Nargesian, F., Zhu, E., Pu, K. Q., & Miller, R. J. (2022). *Data lake management: Challenges and opportunities*. *ACM Computing Surveys*, 55(7), 1–36.
- [7] Oracle. (2025). *Data warehouse and data lake solutions for enterprise analytics*. Oracle Corporation.
- [8] Sakr, S., & Zomaya, A. Y. (2021). *Encyclopedia of big data technologies*. Springer.
- [9] Sawadogo, P., & Darmont, J. (2021). *On data lake architectures and metadata management*. *Journal of Intelligent Information Systems*, 56(1), 97–120.
- [10] Verbitskiy, I., Thiele, M., Lehner, W., & Böhm, A. (2023). *Data lakehouse: A modern architecture for data analytics and business intelligence*. *Information Systems*, 117, 102240